

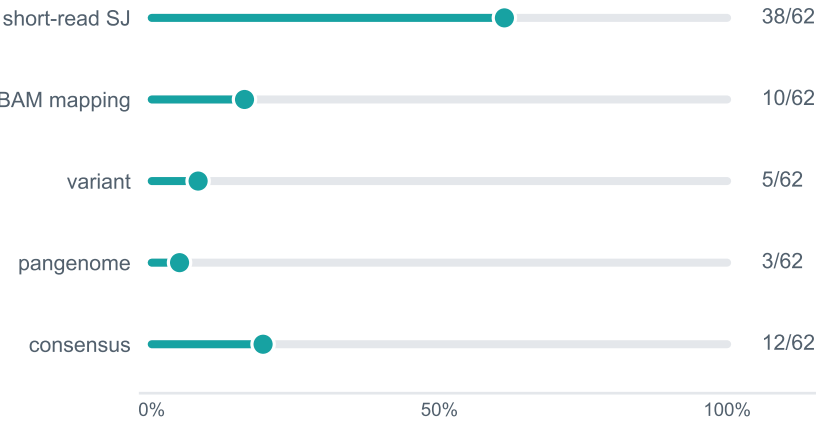
Verdict on 49 novel isoforms

Trustworthy = high/medium-confidence novel · the share you can act on at a glance



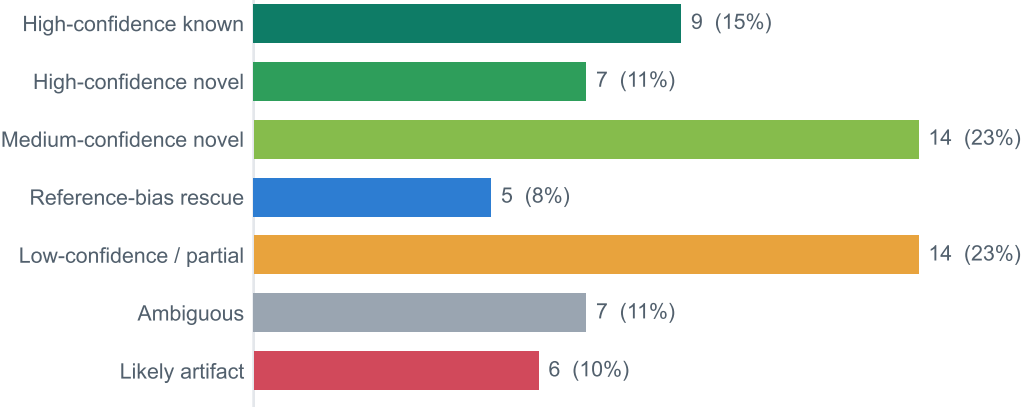
Evidence axes in play

fraction of isoforms for which each axis could be evaluated



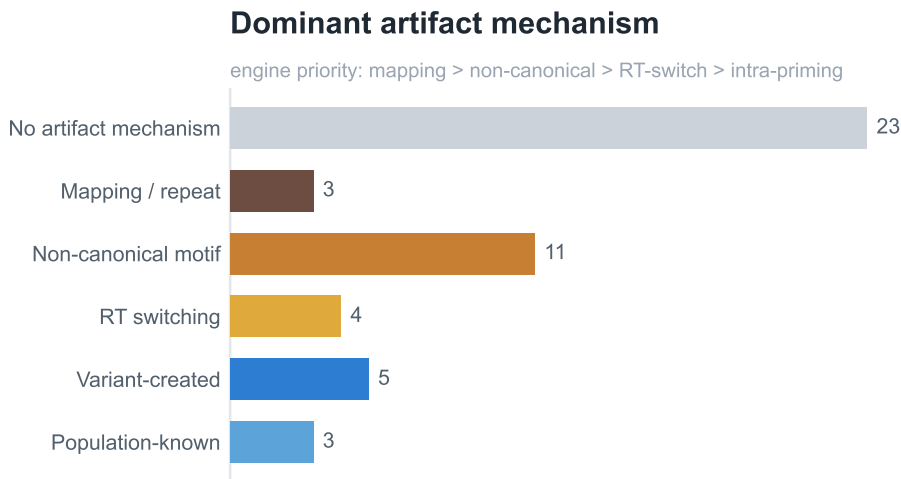
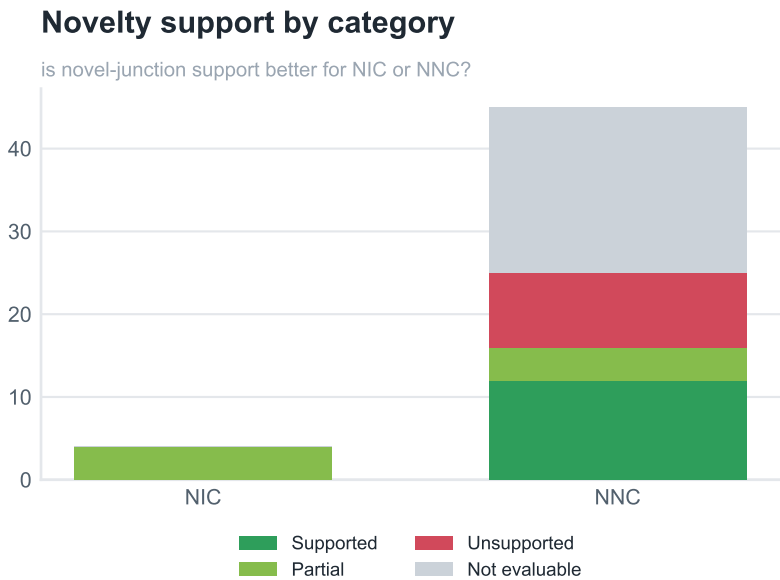
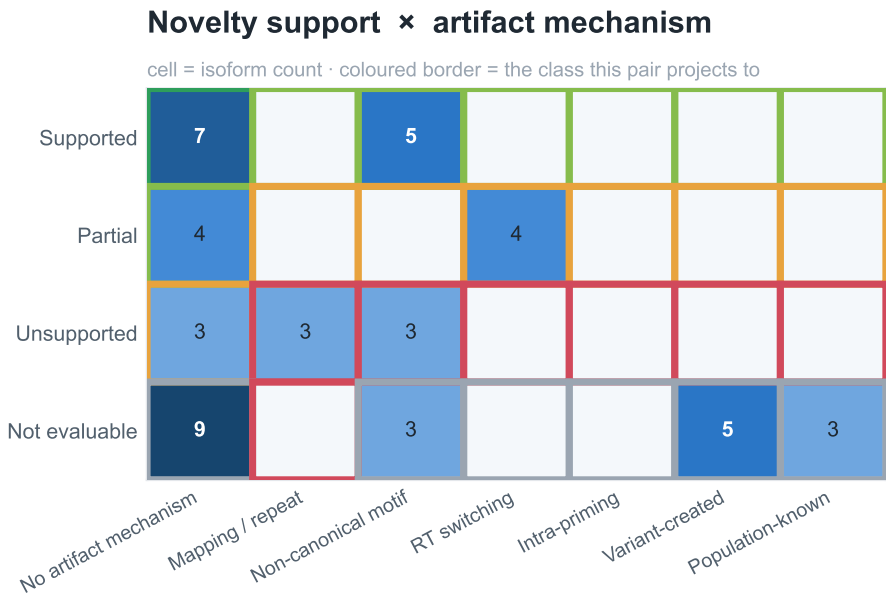
Confidence-class distribution

every isoform · count (share of all)



How each verdict was reached

Every novel isoform is scored on two axes; the pair projects to one confidence class.

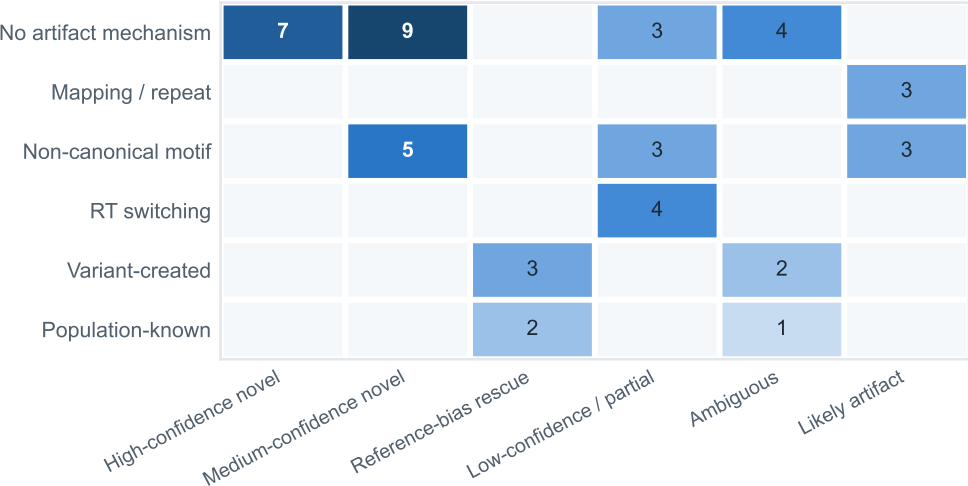


Why calls were down-weighted

Which artifact mechanism drove each verdict, and whether the evidence supports it.

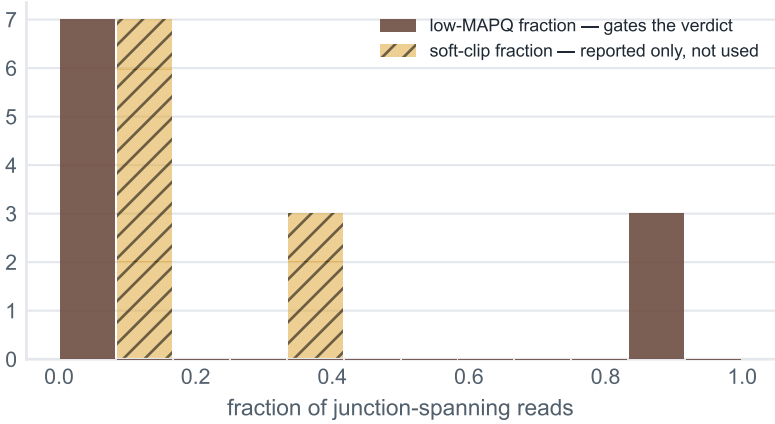
Mechanism → resulting class

where each artifact mechanism ends up



BAM mapping-evidence quality

are mapping artifacts backed by poor alignment?



Structural-signal availability

null is shown as n/a — never silently counted as false

Structural prior	true	false	n/a
Non-canonical motif	11	51	0
RT-switch stage	4	58	0
Caller chain available	62	0	0

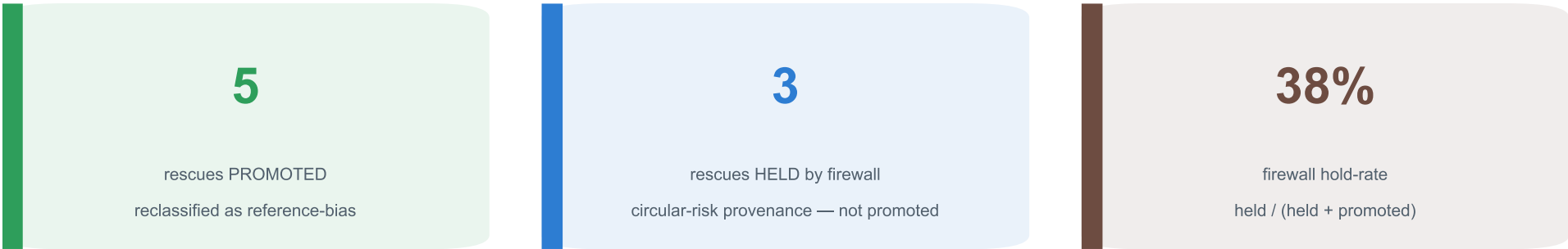
Mapping-down-weighted calls (3)

isoform	class	reads	low-MAPQ	suppl.
NNC_map.36	Likely artifact	14	0.86	0.71
NNC_map.37	Likely artifact	14	0.86	0.71
NNC_map.38	Likely artifact	14	0.86	0.71

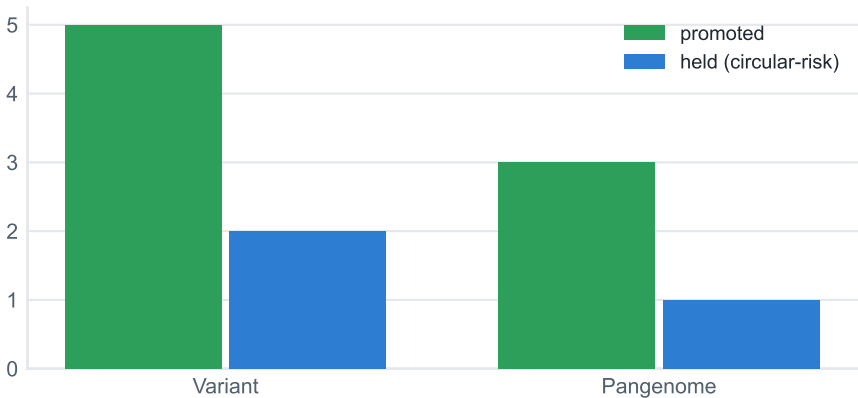
Reference-bias rescue & circularity firewall

A novel junction explained by reference bias is reclassified — but only on independent evidence.

The thesis: nothing is promoted on circular evidence



Rescue axis: promoted vs held



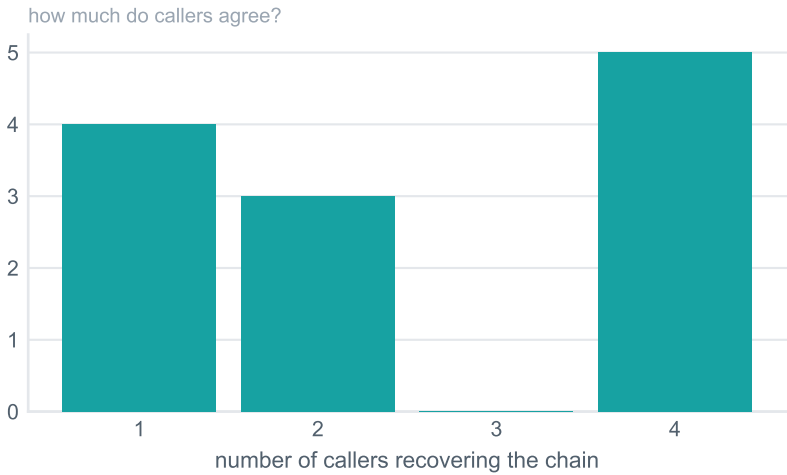
Held-rescue ledger (3)

isoform	category	why held (rule trace)
HELD_var.44	not_in_catalog	variant creates canonical motif on haplotype but prove
HELD_var.45	not_in_catalog	variant creates canonical motif on haplotype but prove
HELD_pan.46	not_in_catalog	all 2/2 novel junctions realizable on a pangenome path

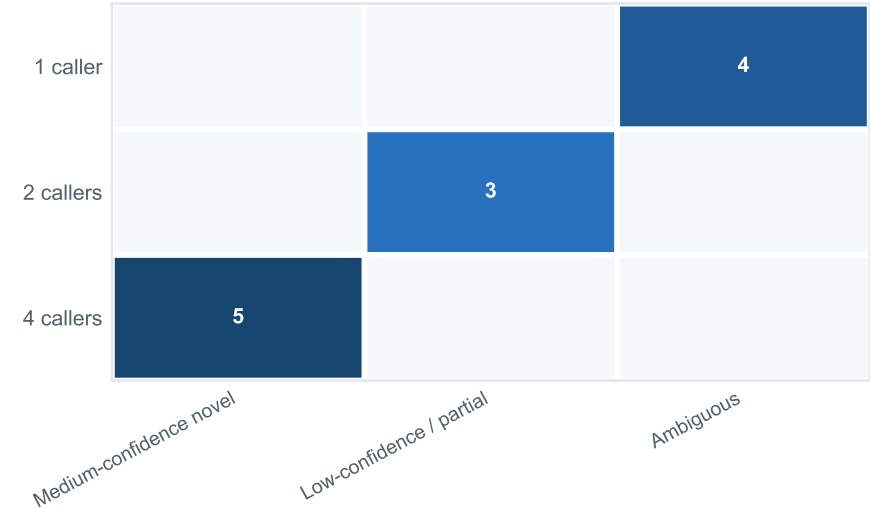
Multi-caller consensus

Caller agreement is methodological corroboration — it promotes to MEDIUM at most, never HIGH.

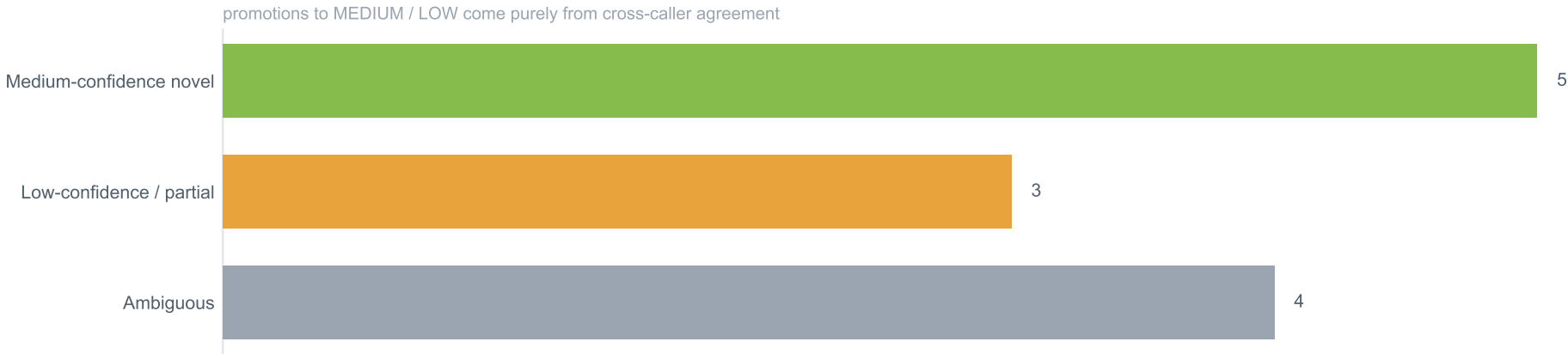
Caller-support distribution



Caller support → confidence class



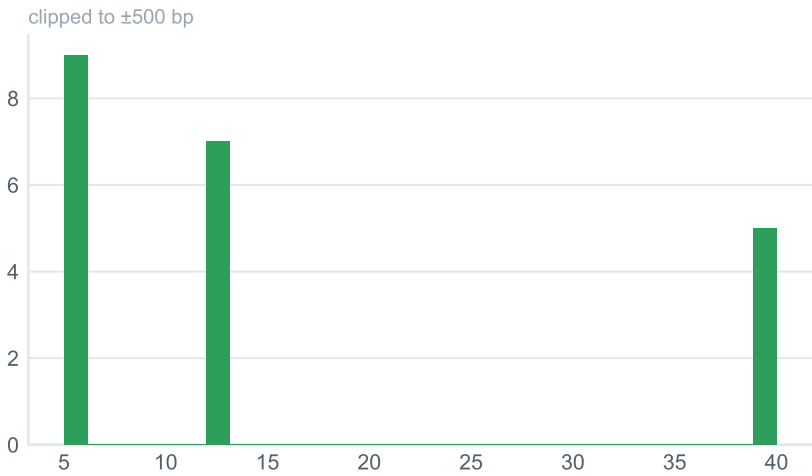
Verdicts of short-read-unknown novels under the consensus axis



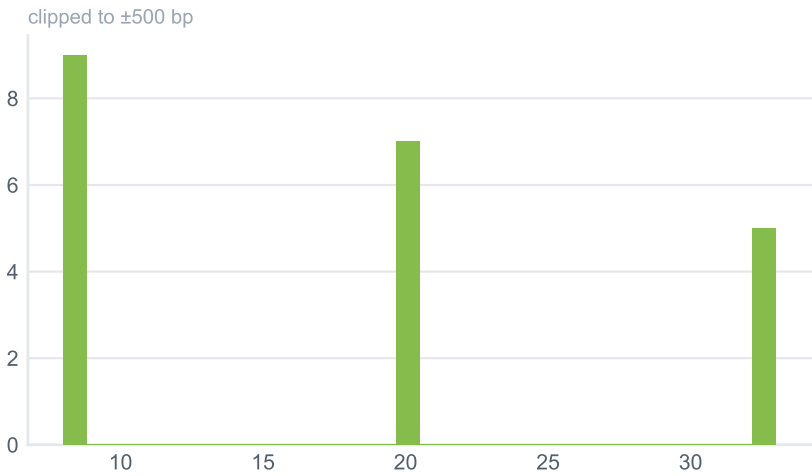
SQANTI3 biological priors

Passed through verbatim — these are NOT used in the PanIsoGuard verdict.

Distance to CAGE peak



Distance to polyA site



Predicted NMD



PolyA motif found

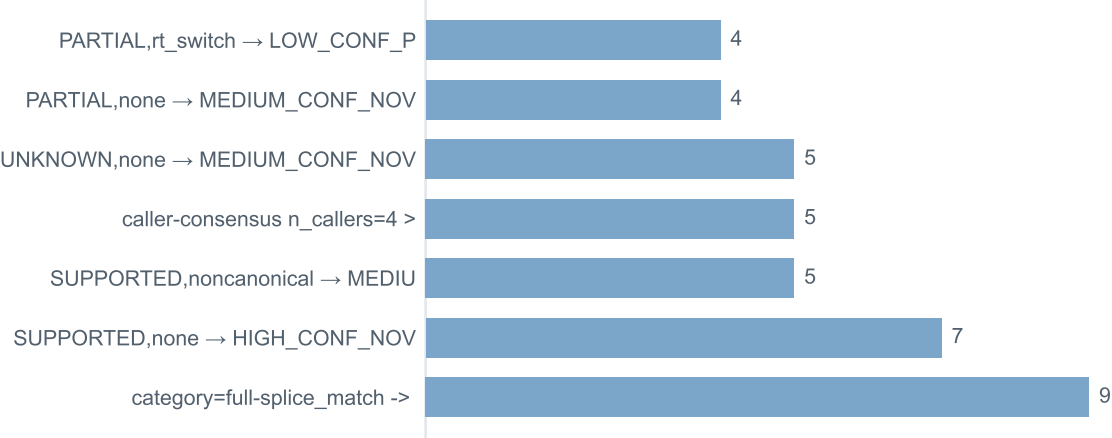


Exactly how this report was produced, and the checks that make it auditable.

Run provenance

key	value
tool_version	0.0.3
ruleset_version	builtin-0.0.1
sqanti3_version_target	6.0
config	<built-in defaults>
classification	showcase (synthetic demo)
isoforms	showcase (synthetic demo)
ref_gtf	showcase (synthetic demo)
sj_tab	showcase (synthetic demo)
bam	showcase (synthetic demo)
axis.short_read	on
axis.catalog	on
axis.bam	on
axis.variant	on
axis.pangenome	on

Most frequent rule-trace lines



Defensibility checklist

- ✓ class tally reconciles (provenance ↔ JSONL)
- ✓ firewall held 3 circular-risk rescue(s)
- consensus axis used this run
- axes in play: short-read, BAM, variant, pangenome, consensus
- SQANTI3 priors consumed, never recomputed